

Abdelrahman Abdelgawad

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Education

Georgia Institute of Technology

Master of Science in Computer Science

Expected Graduation May 2028

Pennsylvania State University

Bachelor of Science in Computer Science

Minor in Mathematics and Minor in Cybersecurity

May 2024

Experience

Undergraduate Research Assistant

Pennsylvania State University

University Park, PA

January 2024 – May 2024

- Collaborated with the professor and teaching assistants on a research paper focused on training visual transformers using machine learning, exploring their mathematical foundations and practical applications.
- Wrote lecture-based questions to enhance student understanding, incorporated them into textbook chapters, and ensured clarity in presenting complex mathematical concepts.
- Authored and reviewed sections of the textbook on advanced topics such as convolutional neural networks, gradient descent, and image classification using MNIST and Fashion-MNIST datasets.
- Reviewed and refined mathematical concepts, equations, and linear algebra applications to ensure technical accuracy and clear presentation in research and teaching materials.

Coding Camp Tutor

Girls Who Code

Brooklyn, NY

June 2023 – August 2023

- Assisted in teaching a crash course on coding fundamentals to 20+ students aged 10-14, covering topics such as variables, conditionals, loops, and event handling.
- Collaborated with camp instructors to develop and structure lesson plans, ensuring they were both engaging and aligned with the camp's educational goals.
- Guided students through the process of designing, creating, and building their own interactive games using Scratch, fostering creativity and problem-solving skills.
- Provided one-on-one support to students, offering clarification on coding concepts or troubleshooting and debugging when they encountered difficulties with Scratch.

Projects

Course Scheduler Interface (Java, MySQL)

- Engineered a robust course scheduling interface, integrating a backend database system for dynamic data storage and retrieval.
- Built an intuitive graphical user interface (GUI) that enables seamless course management, featuring course selection, conflict detection, and schedule visualization.
- Applied error handling techniques to validate user inputs and prevent SQL injections.

FencingTimeClone (React, Google Cloud SQL)

- Developed a personal remake of fencingtimelive.com, a widely used platform for tracking and displaying live fencing tournament results.
- Designed a dynamic UI that allows users to add, modify, and delete fencers and events
- Programmed features for generating and presenting pools and direct elimination (DE) tables, incorporating match result input and real-time ranking updates.

Playing Card Classifier (Python, PyTorch)

- Constructed a deep learning model using PyTorch to classify playing cards with 96% accuracy.
- Automated the training and testing pipeline for a dataset of playing card images, refining model performance through hyperparameter tuning.
- Integrated visualization tools to showcase classification outcomes with clear, easy-to-understand charts and images, enhancing model interpretability.

Skills

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| • Java (Proficient) | • Linux (Intermediate) | • React (Familiar) |
| • Python (Proficient) | • Git (Intermediate) | • NoSQL (Familiar) |
| • SQL (Proficient) | • HTML, CSS (Intermediate) | • Microsoft Office Specialist |
| • C, C++ (Intermediate) | • JavaScript (Intermediate) | Master Certification (2019) |